

Rajenthar Jeganathan

SOFTWARE ENGINEER

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SUMMARY

Accomplished Software Engineer with over six years of experience building robust enterprise applications across aviation systems, real time risk management, and ecommerce platforms. Proficient in Agile, Sprint, and Scrum methodologies, with a strong emphasis on process excellence and team collaboration. Skilled in full stack development, database design, event driven and distributed system design (sagas, CQRS, change data capture, and idempotent transaction processing) and technical documentation, with a proven track record of delivering high quality solutions. Recognised for being self motivated, trustworthy, and highly adaptable in dynamic and collaborative work environments.

CAREER HIGHLIGHTS

- Developed and maintained web applications using Java, Spring, and MySQL.
- Built internal APIs and batch applications leveraging Spring Boot, Scala, Hive, MongoDB, and MySQL.
- Developed Spring Batch jobs for large-scale data processing and transformation.
- Deployed and scheduled batch jobs using Azkaban for efficient pipeline management.
- Created automation scripts and support tools using Shell and Python to streamline development and operations.
- Collaborated in an Agile development environment, using Jira for sprint management and Bitbucket for version control.
- Enhanced code quality by integrating SonarQube and performed performance testing with Taurus.
- Designed and managed database schemas, ensuring data integrity and operational efficiency.
- Authored and reviewed technical documentation, including Proof of Concept (POC) reports and release notes.
- Handled live issue resolution, ensuring consistent application stability and uptime.
- Used tools including Postman, MySQL Workbench, Docker, and Localstack (for local AWS environments) in development and testing workflows.

TECHNICAL SKILLS

Programming	Java, Python, PHP, JavaScript, Scala
Databases	MySQL, MongoDB, Postgres, Hive
Web	React, HTML, CSS, Ag-Grid, Apex-charts, Material UI
Frameworks	Spring Boot, Spring Batch, Spring, Android, Apache Kafka, Redpanda, Redis
Distributed Systems	Event-Driven Architecture, Saga Orchestration, CQRS, Change Data Capture, Transactional Outbox, Idempotency
Testing	JUnit, Mockito, Testcontainers
Version Control	Git (Bitbucket / GitHub)
Cloud	AWS
Tools	Jira, MySQL Workbench, Postman, IntelliJ IDEA, SonarQube, Redis Desktop Manager, Docker, Kubernetes, Jenkins, Kibana, Taurus, Debezium, OpenTelemetry, Grafana, Hadoop toolsets (Sqoop, Beeline, Hive)

CORE COMPETENCIES

- Software and Application Design
- Agile Methodology
- Distributed Systems Design
- Technical Architecture Design and Documentation
- Research and Development
- Event-Driven Architecture

EXPERIENCE

Software Engineer, Rakuten

Jan 2023 – Present

Rakuten is an internet company specialising in high-performance e-commerce applications, mobile networks, banking, and other financial services.

- Performed performance testing using Taurus and Gatling.
- Coordinated end-to-end design, from schema to technical implementation.
- Provided production support for releases, alerts, and schema changes.
- Optimized application performance and implemented graceful shutdown handling.
- Utilized AI tooling to streamline work across the software development lifecycle, with all AI-assisted output independently reviewed and confirmed.
- Authored documentation including release and rollback manuals, API specifications, design documents, and Architecture Decision Records (ADRs).
- Mentored junior engineers and new team members.

Software Engineer, ZeroBeta

Jan 2021 – Dec 2022

Specialised in delivering high-performance computing products and solutions tailored for financial risk management.

- Developed and maintained modern applications using Java, Spring Boot, and MySQL.
- Designed and implemented frontend features with React, Ag-Grid, Apex-charts, Material-UI, and Redux.
- Integrated caching solutions with Redis and implemented message passing via Kafka.
- Participated in Agile sprint planning and version control using Jira and Bitbucket.
- Improved and maintained code quality through SonarQube integration and continuous improvement practices.
- Conducted requirements gathering, POC development, technical feasibility analysis, and performance impact assessments.
- Authored and maintained technical documentation, including database design and data manipulation procedures.
- Provided live issue resolution to support application stability and production environment reliability.

Software Engineer, Information Systems Associates (Accelaero)

Feb 2020 – Jan 2021

Leading provider of advanced technology solutions for the global travel and aviation industry since 2005.

- Developed modern applications using Java, Spring Boot, Protobuf, ANTLR, and PostgreSQL.
- Contributed to frontend development with React and Redux.
- Implemented APIs and business logic, ensuring high JUnit test coverage and maintaining code quality through SonarQube.
- Utilized Jira for sprint management and Bitbucket for version control in an Agile environment.

Software Engineer Intern, GE Transportation Company

Jun 2018 – Dec 2018

Global leader in equipment, systems, digital solutions, and services for freight rail, transit, mining, industrial, and marine sectors.

- Developed and debugged features for products built with Spring and Hibernate.
- Created automation test scripts using Serenity/JS to ensure functional reliability.
- Explored machine learning concepts and implemented a sample K-means clustering algorithm.
- Participated in Agile practices and actively contributed to code reviews to improve product quality.

PERSONAL PROJECTS

Lynx – Multi-Currency Ledger Engine

May 2026 – Present

A distributed ledger system for cross-currency transfers, built to production patterns: saga orchestration, event-driven CQRS, and end-to-end idempotency.

- Designed an append-only, double-entry ledger in PostgreSQL at SERIALIZABLE isolation as the system's single source of truth, with every write balanced and nothing ever updated or deleted.
- Implemented a 3-step saga orchestrator (hold → FX rate lock → settle) with compensating actions on failure, and a recovery worker that resumes orphaned sagas after a crash using FOR UPDATE SKIP LOCKED.

- Guaranteed exactly-once transfers by layering an idempotency key, a deterministic saga identifier, and a database UNIQUE constraint, so retries and crashes cannot produce duplicate ledger entries.
- Published domain events through a transactional outbox streamed off the Postgres write-ahead log by Debezium CDC, ensuring events can never be lost or orphaned from the writes that produced them.
- Built CQRS read projections for balances and transaction history so read traffic never touches the ledger write path.
- Secured service-to-service calls with service-identity tokens carrying on-behalf-of user context, per-client circuit breakers, and correlation IDs traced end-to-end through OpenTelemetry.
- Covered the system with automated tests using JUnit and Testcontainers against a real PostgreSQL instance, and documented each trade-off in Architecture Decision Records.

Java 21 · Spring Boot 3.3 · PostgreSQL · Redis · Kafka/Redpanda · Debezium · OpenTelemetry · Docker · Testcontainers

EDUCATION

BSc. Eng (Hons), Computer Science and Engineering

Nov 2015 – Jan 2020

Faculty of Engineering, University of Moratuwa, Colombo, Sri Lanka